Calculus I: Differentiation

Math 101

Fall 2023

Problem Set 5: Derivatives and Applications

1. Find the derivatives of the following functions:
   (a) $f(x) = 3x^4 - 2x^3 + 5x - 7$
   (b) $g(x) = x^2 \sin(x)$
   (c) $h(x) = \frac{e^x}{1+e^x}$
   (d) $j(x) = \ln(x^2 + 1)$
   (e) $k(x) = (2x + 1)^5 \cdot (3x - 2)^4$
2. Use implicit differentiation to find $\frac{dy}{dx}$ for:
   (a) $x^2 + y^2 = 25$
   (b) $x^3 + y^3 = 6xy$
3. A particle moves along a straight line with position function $s(t) = t^3 - \boxed{6t^2} + 9t$ (in meters), where t is in seconds.
   (a) Find the velocity and acceleration functions.
   (b) When is the particle at rest?
   (c) When is the particle moving backward?
4. Find the equation of the tangent line to the curve $y = x^3 - 3x + 2$ at the point where $x = 2$.
5. A rectangular box with square base is to have a volume of 108 cubic inches. Find the dimensions that minimize the surface area of the box.